

newdb Shared Bundle

External Integration Manual (Expanded)

newdb Project

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1 Document Scope

This manual is an expanded LaTeX edition aligned with `newdb/scripts/examples/README_shared_bundle.md`. It covers:

- Build and distribution of self-contained shared bundles.
- Full exported interface references (`newdb` and `gtest_capi`).
- Command-driven external integration model.
- Unified result parsing and error handling strategy.
- Practical cross-language integration notes.

2 Shared Bundle Delivery Model

2.1 Output directories

- VS/MSVC: `build/newdb-vs2026-mt-tests/shared_bundle/Release`
- MinGW: `build/newdb-mingw/shared_bundle/Release`
- WSL/Linux: `build/newdb-wsl-tests/shared_bundle/Release`

2.2 Build commands

- VS:

```
cmake --build build/newdb-vs2026-mt-tests --config Release --target shared_bundle --parallel
```
- MinGW:

```
cmake --build build/newdb-mingw --target shared_bundle --parallel
```
- WSL:

```
cmake --build build/newdb-wsl-tests --target shared_bundle -j8
```

2.3 Dependency precedence strategy

- Windows: `run_*.cmd` launchers prioritize bundle-local DLL lookup.
- Linux/WSL: binaries are linked with `RPATH=$ORIGIN`, prioritizing colocated `.so`.
- MinGW: runtime DLLs (`libstdc++-6.dll`, `libgcc_s_*.dll`, `libwinpthread-1.dll`) are bundled.

3 newdb C API Full Reference

3.1 Header and constants

Header: `newdb/engine/include/newdb/c_api.h`

- API version: `NEWDB_C_API_VERSION_MAJOR/MINOR/PATCH`
- ABI version: `NEWDB_C_API_ABI_VERSION`

3.2 Error codes

Code	Meaning
NEWDB_OK	Success
NEWDB_ERR_INVALID_ARGUMENT	Invalid argument or buffer contract violation
NEWDB_ERR_INVALID_HANDLE	Invalid or already-destroyed session handle
NEWDB_ERR_EXECUTION_FAILED	Command/business execution failed
NEWDB_ERR_INTERNAL	Internal engine failure
NEWDB_ERR_LOG_IO	File/log I/O failure
NEWDB_ERR_SESSION_TERMINATED	Session is terminated and cannot continue

3.3 Thread-safety contract

- Global metadata APIs are thread-safe.
- Per-session APIs are safe only under “one handle per thread”.
- Sharing one handle across concurrent threads is not guaranteed safe.

3.4 Function catalog

Version and ABI

- `newdb_version_string`
- `newdb_api_version_major/minor/patch`
- `newdb_abi_version`
- `newdb_negotiate_abi`
- `newdb_error_code_string`

Utility

- `newdb_sum`
- `newdb_check_schema_file`

Session lifecycle

- `newdb_session_create`
- `newdb_session_destroy`
- `newdb_session_set_table`
- `newdb_session_last_error`

Execution and telemetry

- `newdb_session_execute`
- `newdb_session_runtime_stats`
- `newdb_session_append_runtime_snapshot`

3.5 Recommended lifecycle

1. Check ABI compatibility with `newdb_negotiate_abi`.
2. Create session handle.
3. Execute command stream via `newdb_session_execute`.
4. Collect stats/snapshots as needed.
5. Destroy session.

4 gtest_capi Full Reference

Header: `newdb/include/gtest_capi.h`

4.1 Capabilities

- Initialize and run GoogleTest from external language runtimes.
- Enumerate tests and fetch test metadata.
- Configure runtime options (filter/output/repeat/shuffle/seed/etc.).
- Read post-run counters and elapsed time.

4.2 Main APIs

- Init and run: `gtest_capi_init_from_argv`, `gtest_capi_run_all`
- Listing and metadata: `gtest_capi_list_tests`, `gtest_capi_enumerate_tests`
- Options: `set/get_filter`, `set/get_output`, `set/get_repeat`, `set/get_shuffle`, `set/get_random_seed`, `set/get_color`, `set/get_break_on_failure`, `set/get_throw_on_failure`, `set/get_catch_exceptions`, `set/get_also_run_disabled_tests`, `set/get_brief`, `set/get_print_time`
- Generic option bridge: `gtest_capi_set_option`, `gtest_capi_get_option`
- Counters: total suite/test, to-run count, passed/failed/skipped/disabled, reportable disabled, elapsed ms.

5 Feature-Oriented External Integration

From external systems, `newdb` exposes full engine capability through command execution:

- DDL/schema management.
- DML and query paths.
- Transaction, conflict, and isolation controls.
- WAL/recovery controls.
- Runtime and maintenance controls (vacuum/group commit/hot index).

5.1 Recommended wrapper layering

- DDL wrapper
- DML wrapper
- Query wrapper
- Txn wrapper
- Runtime tuning wrapper

6 Command Templates

6.1 DDL

```
CREATE TABLE(users)
USE(users)
SET PRIMARY KEY(uid)
CREATE SCHEMA(finance_v1)
ALTER TABLE users SET SCHEMA(finance_v1)
RENAME TABLE(users_v2)
DROP TABLE(users_v2)
```

6.2 DML and query

```
INSERT(1, alice, 1000, rd)
UPDATE(1, alice, 1200)
DELETE(1)
COUNT
WHERE(balance, >, 1000, AND, dept, =, rd)
PAGE(1, 20, id, desc)
SUM(balance)
AVG(balance)
```

6.3 Transaction and tuning

```
BEGIN
SAVEPOINT(sp1)
ROLLBACK TO(sp1)
COMMIT
TXNISOLATION snapshot
WRITECONFLICT wait 2000
AUTOVACUUM threshold 500
WALSYNC normal 200
GROUPCOMMIT 10 128
```

7 Result Parsing and Error Handling

Always evaluate all three:

1. rc from newdb_session_execute
2. newdb_session_last_error
3. output_buf

7.1 Recommended result model

```
ExecuteResult {  
    rc: int  
    last_error: int  
    error_name: string  
    output: string  
    ok: bool  
}
```

7.2 Buffer sizing

Use at least 4KB for simple commands and 16KB+ for complex query outputs. Do not rely on text completeness when truncation is possible.

8 Cross-language Practical Notes

8.1 Python/C#/Node strategy

- Load main library via absolute path.
- Keep one session handle per worker/thread.
- Wrap command execution in one adapter method returning unified result object.
- Convert non-NEWDB_OK into domain-specific exceptions.

9 Operational Guidance

9.1 Telemetry

Use:

- `newdb_session_runtime_stats` for real-time metric snapshots.
- `newdb_session_append_runtime_snapshot` for JSONL historical traces.

9.2 Suggested key metrics

- Commit latency (p95/max)
- Write conflict and timeout counts
- WAL recovery elapsed and scanned records
- Vacuum trigger count and bytes reclaimed

10 Distribution Checklist

1. Ship entire `shared_bundle/Release` directory.
2. Do not split individual DLL/SO from their runtime dependencies.
3. Enforce absolute-path dynamic loading in host app.
4. Validate ABI before first command execution.
5. Persist command-level execution records.

11 Troubleshooting

- “Module not found”: missing colocated dependencies.
- Unexpected library resolution: enforce absolute load and bundle launchers.
- WSL googletest download failures: use local `FETCHCONTENT_SOURCE_DIR_GOOGLETEST`.